

Effect of Inorganic Sulfate and Sulfur Amino Acids on Bone Mineralization in Broilers¹

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ABSTRACT Two experiments with a factorial arrangement of treatments were conducted to determine the effect of inorganic sulfate and sulfur amino acids (SAA) levels on bone mineralization and Ca excretion in broilers from 1 to 21 days of age. Experiment 1 included two levels of Ca (.6 and 1.0%), two sulfate sources [copper sulfate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), and potassium sulfate (K_2SO_4)], and three sulfate levels (.024, .034, and .044%). Experiment 2 included two Ca levels (.6 and 1.0%), two CP levels (18 and 22%), and three SAA levels (.93, 1.23, and 1.53%). Body weight gain, feed:gain, feed consumption, and tibia variables were measured in both experiments. In Experiment 2, Ca retention was also assessed. In Experiment 1, birds fed potassium sulfate and .6% Ca gained significantly more weight than birds fed copper sulfate and 1.0% Ca, respectively. As sulfate level increased, tibia ash percentage increased and decreased linearly in birds fed copper sulfate and potassium sulfate, respectively ($P < .012$). In Experiment 2, birds fed 1.0% Ca, .93% SAA, and 22% CP performed better than the birds receiving the other levels. A low Ca level (.6%) was more detrimental for bone mineralization in birds fed 22% CP compared with birds fed 18% CP. There was no effect of SAA on tibia ash and Ca percentages. Fecal Ca excretion decreased linearly with SAA level ($P < .01$). The results suggest that inorganic sulfate from the potassium salt may adversely affect Ca deposition in bone. However, SAA supplementation up to 65% in excess of the requirement in starter broiler diets did not affect bone mineralization. (*Key words:* sulfate, sulfur amino acids, tibia ash, calcium, protein)

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INTRODUCTION

In spite of much effort, leg weakness continues to be a significant problem for the commercial poultry industry. Sulfate salts, including calcium sulfate, potassium sulfate, magnesium sulfate, and sodium sulfate, are utilized in broiler diets to meet mineral requirements. Copper sulfate is often used as a growth promotant (Scott *et al.*, 1982). However, limited attention has been given to the effects of sulfate from

these supplements on bone mineralization in poultry. In mammals, it is well known that excess dietary protein and inorganic sulfate increase urinary Ca (Whiting and Draper, 1980, 1981; Zemel *et al.*, 1981; Greger *et al.*, 1991) and appear to have an adverse effect on bone.

In poultry, excess dietary protein and inorganic sulfate have been associated with increased incidence of leg abnormalities. Stevens and Salmon (1988) observed a linear increase in the incidence of rotated tibia of turkey poults at 2 wk of age when fed protein levels from 24 to 39%. Moreover, an increased incidence of valgus-varus deformation and lower bone ash percentage were observed when turkeys were fed a high-protein diet from 3 to 20 wk of age (Stevens and Salmon, 1988).

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Because protein level did not significantly affect BW gain at 20 wk, the authors hypothesized that the acid load *per se* due to the increase in catabolism of sulfur amino acids (SAA) may have induced the leg abnormalities. Stevens and Salmon (1989) also determined that turkey poults fed from 0 to 4 wk of age diets containing .558% DL-Met supplemented with .327% of sulfate as calcium sulfate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) and magnesium sulfate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$) had an increased incidence of leg abnormalities, especially at marginal dietary levels of Ca and P.

Using broiler chickens, Skinner *et al.* (1991) observed that increasing the level of total amino acids up to 120% of the requirement significantly decreased percentage tibia ash in birds fed .5 and 1.0% Ca diets; however, the decline in tibia ash was more severe for birds fed .5% Ca. Halley *et al.* (1987) observed an increased incidence of varus deformation in broilers fed 1.8% sulfate from 1 day to 3 wk of age, but no effect on other variables such as bone ash. Edwards (1984) observed that although .31% sulfate in broiler diets significantly increased bone ash percentage over the control, the incidence of tibial dyschondroplasia (TD) was not affected.

Unfortunately, all of these studies were conducted using levels of sulfate that greatly exceed the levels found in practical broiler diets. To date, experiments have not been performed to study the effect of excess SAA increasing total sulfur or acid load on bone mineralization in broilers, even though liberal use of SAA is commonly practiced in formulating commercial diets. The objective of this study was to determine the effect of inorganic sulfate level derived from different sources, protein level, and excess SAA on bone mineralization in broilers.

MATERIALS AND METHODS

Experiment 1

Four hundred and thirty-two Cobb $\times$ Cobb 1-day-old broiler chicks were ran-

domly assigned in a $2 \times 2 \times 3$ factorial arrangement of treatments. Each treatment was replicated six times with six birds per replicate pen. The factors involved in this study were two sulfate sources [copper sulfate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) and potassium sulfate (K_2SO_4)], two Ca levels (.6 and 1.0%), and three sulfate levels (.024, .034, and .044%). Composition of the basal diets is presented in Table 1. The levels of Ca were obtained by using different levels of ground limestone. The levels of sulfate were obtained by supplementing the basal diets with .025, .050, and .075% of copper sulfate (38.47% SO_4 and 25.46% Cu) and with .0175, .035, and .0525% of potassium sulfate (55% SO_4) in place of corn. The levels of sulfate chosen were based on a maximum of 190 ppm of Cu in the experimental diet coming from the highest level of copper sulfate supplementation. All diets were formulated to meet or exceed the nutrient requirement guidelines of the NRC (1984).

Birds were maintained in electrically heated, Petersime³ starting batteries with raised wire floors from 1 to 21 days of age. Experimental diets and water were provided for *ad libitum* consumption. Body weight and feed consumption of each replicate group were recorded when chicks were 14 and 21 days of age. At 20 days the birds were checked individually for the presence of any leg weakness. The experiment was terminated when the birds were 21 days old. At this time, three randomly selected chicks from each replicate were euthanatized by cervical dislocation, and the left tibia of each bird was removed. After being placed in boiling water for 3 min, tibias were cleaned of muscle and connective tissue. Samples were dried at 105 C for 24 h, extracted with diethyl ether for 48 h, weighed, and ashed at 600 C for 24 h. After cooling, the bone ash was weighed, ash percentage calculated, and the bone ash was ground with a mortar and pestle. Calcium content in feeds and bone ash samples were analyzed by flame atomic absorption spectroscopy.⁴

Experiment 2

In this study, 432 Cobb $\times$ Cobb 1-day-old broiler chicks were randomly assigned in a $2 \times 2 \times 3$ factorial arrangement

³Petersime Co., Gettysburg, OH 45328.

⁴Varian Techtron Pty. Ltd., Mulgrave, Victoria, Australia 3170.

TABLE 1. Composition of basal diets, Experiment 1¹

Ingredients and composition	.6% Ca	1.0% Ca
	(%)	
Ground yellow corn	52.9	51.8
Soybean meal (47.5% CP)	36.4	36.4
Animal tallow	5.5	5.5
Alfalfa, dehydrated (17% CP)	1.0	1.0
DL-methionine (99%)	.2	.2
Monocalcium phosphate	2.0	2.0
Ground limestone	.36	1.41
Sodium chloride	.40	.40
Premix ²	1.20	1.20
Flavomycin ³	.05	.05
Calculated composition		
ME, kcal/kg	3,139	3,104
Protein	22	22
Calcium	.6	1.0
Phosphorus (available)	.5	.5
Methionine and cystine	.91	.91
Sulfate	.01	.01

¹The different levels of sulfate (.024, .034, and .044%) were obtained by supplementing the basal diets with .025, .050, and .075% of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (38.47% SO_4 and 25.46% Cu) or with .0175, .035, and .0525% K_2SO_4 (55% SO_4) in place of corn.

²Premix furnished the following per kilogram of diet: vitamin A, 11,000 IU (retinylacetate); cholecalciferol, 4,400 ICU; menadione, sodium bisulfite, 3.3 mg; vitamin E, 11 IU (dl, α -tocopherol acetate); riboflavin, 6.6 mg; calcium pantothenate, 11.9 mg; niacin, 77 mg; d-biotin, .11 mg; folacin, .66 mg; B₁₂, .011 mg; choline chloride, 880 mg; copper, 10 mg; iron, 100 mg; manganese, 100 mg; zinc, 100 mg; iodine, 3.0 mg; cobalt, 1.0 mg; calcium, 92 mg; and selenium, .2 mg.

³Hoechst-Roussel Agri-Vet Co., Somerville, NJ 08876; provided 2.2 ppm bambarmycins in the diet.

of treatments. Each treatment was replicated six times with six chicks per replicate pen. The treatment arrangement included two CP levels (18 and 22%), two Ca levels (.6 and 1.0%), and three SAA levels (.93, 1.23, and 1.53%). Four corn-soybean meal basal diets were formulated to achieve the different protein and Ca levels (Table 2). The levels of SAA were obtained by supplementing the basal diets (.93% SAA) with .30 and .60% DL-Met in place of corn. The 18% CP diet was supplemented with synthetic L-Arg-HCl and feed-grade L-Lys-HCl and L-Thr, so that all essential amino acids met or exceeded the NRC (1984) guidelines. This experiment was also conducted from 1 to 21 days of age, and the management of the birds was the same as described for Experiment 1. Body weight and feed consumption of each replicate pen were recorded at 14 and 21 days. Total fecal output was collected for 48 h from Days 18 to 20 to determine the treatment effect on Ca excretion. Feces from each replicate were collected on wax paper and, after

removal of feather and feed contamination, were freeze-dried and ashed for further analysis. Feed consumption of replicate pens was recorded at the same time. The experiment was terminated when the birds were 21 days old. The protocol for collection, preparation, and analysis of tibia samples was the same as described for Experiment 1. As in Experiment 1, flame atomic absorption spectroscopy⁴ was used to determine Ca content in feed, tibia, and feces. Percentage Ca excretion was calculated as grams of Ca in feces divided by grams of Ca consumed in the 48-h period.

Statistical Analysis

All data were analyzed by ANOVA procedure appropriate for a factorial arrangement of treatments in a completely randomized design using the General Linear Models procedure of SAS[®] software (SAS Institute, 1985). Duncan's multiple range test and orthogonal contrasts were used to compare treatment means (Steel and Torrie, 1980).

TABLE 2. Composition of basal diets, Experiment 2¹

Ingredients and composition	18% CP .6% Ca	18% CP 1.0% Ca	22% CP .6% Ca	22% CP 1.0% Ca
	(%)			
Ground yellow corn	64.8	62.4	54.6	52.7
Soybean meal (47.5% CP)	26.3	26.3	36.6	36.6
Animal tallow	4.0	4.8	5.6	6.4
Ground limestone	.38	1.44	.35	1.40
Monocalcium phosphate	2.00	2.00	1.90	1.90
DL-methionine (99%)	.34	.34	.23	.23
L-threonine	.07	.07	. . .	. . .
L-lysine	.30	.30	. . .	. . .
L-arginine	.19	.19	. . .	. . .
Sodium chloride	.4	.4	.4	.4
Premix ²	1.20	1.20	1.20	1.20
Flavomycin ³	.05	.05	.05	.05
Calculated composition				
ME, kcal/kg	3,120	3,119	3,153	3,151
Protein	18	18	22	22
Calcium	.6	1.0	.6	1.0
Phosphorus (available)	.5	.5	.5	.5
Methionine and cystine	.93	.93	.93	.93
Lysine	1.29	1.29	1.29	1.29
Arginine	1.45	1.45	1.58	1.58
Threonine	.80	.80	.89	.89

¹The different levels and SAA (.93, 1.23, and 1.53%) were obtained by supplementing the basal diets (.93% SAA) with .30 and .60% DL-methionine in place of corn.

²Premix furnished the following per kilogram of diet: vitamin A, 11,000 IU (retinylacetate); cholecalciferol, 4,400 ICU; menadione, sodium bisulfite, 3.3 mg; vitamin E, 11 IU (dl,α-tocopherol acetate); riboflavin, 6.6 mg; calcium pantothenate, 11.9 mg; niacin, 77 mg; d-biotin, .11 mg; folacin, .66 mg; B₁₂, .011 mg; choline chloride, 880 mg; copper, 10 mg; iron, 100 mg; manganese, 100 mg; zinc, 100 mg; iodine, 3.0 mg; cobalt, 1.0 mg; calcium, 92 mg; and selenium, .2 mg.

³Hoechst-Roussel Agri-Vet Co., Somerville, NJ 08876; provided 2.2 ppm bambermycins in the diet.

RESULTS

Experiment 1

Only main effect treatment means for BW gain, feed:gain, and feed consumption at 21 days are presented because no significant interactions were observed (Table 3). Birds receiving potassium sulfate had significantly greater BW gain at 21 days than those receiving copper sulfate. The source of sulfate did not affect feed:gain at 21 days. Birds fed a marginal Ca level (.6%) gained significantly more weight and had better feed:gain at 21 days compared with birds fed 1.0% Ca. Body weight gain and feed:gain were not significantly affected by dietary sulfate level at 21 days of age. None of the treatments affected feed consumption.

The effect of sulfate source, and Ca, and sulfate level on tibia variables and leg

weakness are presented in Table 4. As expected, percentage tibia ash and percentage tibia Ca were significantly greater for birds fed 1.0% Ca in relation to birds fed .6% Ca. There was a significant interaction between sulfate source and sulfate level for percentage tibia ash. Tibia ash increased linearly when increasing levels of sulfate were given in birds fed copper sulfate. However, when birds were fed potassium sulfate, tibia ash decreased linearly with the level of sulfate ($P = .011$). Increasing the level of sulfate in the diet decreased percentage tibia Ca ($P = .067$), but no significant interactions were observed for this variable.

There were no significant effects of sulfate source on percentage tibia Ca. Tibia dry weight and the incidence of weak legs were not significantly affected by the different dietary treatments, and no significant interactions were observed.

TABLE 3. Effect of sulfate source, calcium level, and sulfate level on body weight gain, feed consumption, and feed:gain from 0 to 21 days of age, Experiment 1^{1,2}

Factor	Body weight gain	Feed consumption	Feed:gain
	(g)	(g)	(g:g)
Sulfate source			
K ₂ SO ₄	612 ^a	862	1.46
CuSO ₄ ·5H ₂ O	594 ^b	849	1.46
Calcium, %			
.6	614 ^a	860	1.44 ^b
1.0	592 ^b	852	1.48 ^a
Pooled SEM ³	6.3	13.4	.008
Sulfate level (SL), %			
.024	612	871	1.46
.034	599	852	1.46
.044	599	845	1.47
Pooled SEM ⁴	7.8	16.4	.009
Contrasts		Probability	
SL linear	.228	.273	.276
SL quadratic	.458	.772	.705

^{a,b}Values of the same variable within factor with no common superscripts differ significantly ($P < .05$).

¹Only main effect means are shown because no significant interactions were observed.

²Each value represents the average of six birds per replication and six replications per treatment.

³Based on 36 observations per mean.

⁴Based on 24 observations per mean.

Experiment 2

Body weight gain, feed consumption, and feed:gain data are presented in Table 5. Body weight gains at 21 days were significantly greater and feed:gain significantly better for birds fed 22% CP than for birds fed 18% CP. Moreover, birds fed 1.0% Ca gained significantly more weight and consumed significantly more feed than birds fed .6% Ca. Increasing the level of SAA linearly decreased BW gain and feed consumption at 21 days. A significant CP by SAA interaction was observed for feed:gain at 21 days. When birds were fed 18% CP, increasing the level of SAA supplementation was detrimental for feed:gain. However, increasing the level of SAA did not affect feed:gain of birds fed 22% CP.

The effect of treatment on tibia variables and Ca excretion are presented in Table 6. A Ca by CP level interaction ($P = .054$) was observed for tibia ash percentage. Tibia ash was lower in birds fed .6% Ca compared with birds fed 1.0% Ca. However, tibia ash decreased more with Ca level when birds were fed 22% CP. There was no effect of SAA on percentage tibia ash. Percentage

tibia Ca was not significantly affected by either SAA, CP, or Ca level.

Tibia dry weight was significantly greater for birds fed 1.0% Ca compared with birds fed .6% Ca. Increasing the level of SAA decreased tibia dry weight linearly. However, neither an effect of protein nor significant interactions were observed for tibia dry weight. A significant Ca by SAA interaction was observed for percentage Ca excretion. As SAA level increased in birds fed .6% Ca, Ca excretion did not change significantly. However, when birds were fed 1.0% Ca, Ca excretion had a quadratic ($P < .003$, not shown) response to SAA level.

DISCUSSION

High levels of Cu (125 to 250 ppm) as copper sulfate are often added to poultry diets as a growth stimulant and as an antifungal agent (Leach *et al.*, 1990). According to Robbins and Baker (1980), 250 ppm of Cu in the diet of immature chickens added as copper sulfate produces gizzard erosion and reduces cysteine utilization and growth. In the present study, the levels of Cu in the diets supplemented

TABLE 4. Effect of sulfate source, calcium level, and sulfate level on tibia parameters and incidence of leg weakness, Experiment 1

Sulfate source	Treatment		Tibia			Leg weakness ⁴
	Ca	Sulfate	Ca ¹	Ash ²	TDW ³	
	(%)	(%)	(%)	(%)	(g)	(%)
CuSO ₄	1.0	.024	35.5 ^a	44.0 ^{ab}	2.05	8.3
	1.0	.034	34.7 ^{ab}	44.3 ^a	2.11	8.9
	1.0	.044	35.4 ^{ab}	44.4 ^a	2.39	30.6
	.6	.024	35.1 ^{ab}	41.2 ^e	2.05	13.9
	.6	.034	34.4 ^{ab}	42.8 ^{bcd}	1.96	11.1
	.6	.044	34.3 ^b	42.4 ^{cde}	1.93	13.9
K ₂ SO ₄	1.0	.024	35.3 ^{ab}	44.4 ^a	2.11	22.8
	1.0	.034	35.0 ^{ab}	44.2 ^a	2.08	13.9
	1.0	.044	35.0 ^{ab}	43.0 ^{bc}	2.19	15.6
	.6	.024	34.8 ^{ab}	42.1 ^{cde}	2.33	20.8
	.6	.034	34.7 ^{ab}	41.7 ^{de}	2.22	26.7
	.6	.044	34.4 ^{ab}	42.2 ^{cde}	2.00	26.1
Pooled SEM			.33	.40	.15	6.7
Source of variation	df			Probability		
Sulfate source (SS)	1		.856	.254	.410	.099
Ca	1		.008	<.001	.413	.594
Sulfate level (SL)	2		.078	.517	.140	.200
SS × Ca	1		.750	.586	.922	.374
SS × SL	2		.540	.023	.536	.362
Ca × SL	2		.567	.141	.124	.545
SS × Ca × SL	2		.747	.122	.986	.199
Contrasts						
SL linear	1		.067	.761	.957	.292
SL quadratic	1		.182	.269	.690	.354
SS × SL linear	1		.826	.011	.287	.210
SS × SL quadratic	1		.279	.282	.747	.499

^{a-e}Values within the same column with no common superscripts differ significantly ($P < .05$).

¹Average percentage tibia calcium of three birds per replication and six replications per treatment.

²Average tibia ash percentage of three birds per replication and six replications per treatment.

³Average weight of dried and fat-free tibia removed from three birds per replication.

⁴Number of birds with leg weakness at 20 days over the total number of birds per replication.

with copper sulfate were 63, 127, and 190 ppm for birds fed .024, .034, and .044% sulfate, respectively. Therefore, it is difficult to conclude that Cu was responsible for the significantly lower BW gain of birds fed copper sulfate. Although not significant, the lower feed consumption of birds supplemented with copper sulfate may in part explain their lower BW gain.

Although .6% Ca is not considered a deficient level for growing broilers (Scott *et al.*, 1982), the better performance of these birds compared with the ones fed 1.0% Ca in Experiment 1 was unexpected. Furthermore, the analyzed Ca content of the diets was very close to the calculated values. A possible explanation for this

difference might be the slightly higher energy level of the .6% Ca diets.

Feeding sulfate salts to mammals is known to increase acid excretion, which is accompanied by hypercalciuria and an increase in bone resorption (Whiting and Draper, 1980, 1981). This effect of sulfate on Ca metabolism may explain the decrease in tibia ash as sulfate level increased in birds fed potassium sulfate. Sulfate anion can exert its effect through changes in blood acid-base balance toward a lower pH and, therefore, increase acid excretion (Zemel *et al.*, 1981; Halley *et al.*, 1987); however, the response appears to vary depending on the source of sulfate. Edwards (1984) reported that excess sul-

TABLE 5. Effect of protein level, calcium level, and sulfur amino acid levels on body weight gain, feed consumption, and feed:gain from 0 to 21 days of age, Experiment 2¹

Protein (P)	Ca	SAA	Body weight gain	Feed consumption	Feed:gain
	(%)		(g)		(g:g)
18	1.0	.93	631 ^{ab}	971 ^a	1.54 ^{ab}
18	1.0	1.23	616 ^{ab}	919 ^{abc}	1.50 ^{abc}
18	1.0	1.53	590 ^{bc}	917 ^{abc}	1.56 ^a
18	.6	.93	606 ^{ab}	882 ^{bcd}	1.45 ^{bcd}
18	.6	1.23	602 ^{ab}	896 ^{bcd}	1.49 ^{abc}
18	.6	1.53	540 ^c	851 ^{cd}	1.58 ^a
22	1.0	.93	654 ^a	927 ^{ab}	1.42 ^{cd}
22	1.0	1.23	659 ^a	949 ^{ab}	1.42 ^{cd}
22	1.0	1.53	609 ^{ab}	847 ^{cd}	1.39 ^d
22	.6	.93	632 ^{ab}	906 ^{abcd}	1.43 ^{cd}
22	.6	1.23	588 ^{bc}	844 ^d	1.44 ^{cd}
22	.6	1.53	606 ^{ab}	853 ^{cd}	1.41 ^{cd}
Pooled SEM			17.3	22.3	.03
Source of variation	df		Probability		
P	1		.009	.158	<.001
Ca	1		.003	<.001	.866
SAA	2		.002	.004	.416
P × Ca	1		.897	.449	.866
P × SAA	2		.521	.683	.014
Ca × SAA	2		.701	.527	.395
P × Ca × SAA	2		.111	.025	.443
Contrasts					
SAA linear	1		<.001	.001	.233
SAA quadratic	1		.478	.574	.567

^{a-d}Values within the same column with no common superscripts differ significantly ($P < .05$).

¹Each value represents the average of six birds per replication and six replications per treatment.

fate as sodium sulfate in broiler diets increased bone ash and had no effect on the incidence of TD. However, Halley *et al.* (1987) observed a higher incidence of TD in broiler chickens fed high levels of ammonium sulfate. The significant sulfate source by sulfate level interaction for percentage tibia ash observed in the present study suggests that the sulfate anions contained in copper sulfate and potassium sulfate may behave differently in their effect on Ca metabolism.

Results from the second experiment show that although all essential amino acids were fed at adequate levels, the chicks fed 18% CP had lower BW gain and feed:gain as compared with those fed 22% CP. These results agree with the findings of others (Fancher and Jensen, 1989).

The detrimental effect of excess SAA on feed intake and performance is more likely a consequence of amino acid imbalance. The imbalance was stronger in birds fed 18% CP compared with birds fed

22% CP. The 18% CP diet was supplemented with Lys, Arg, and Thr to meet or exceed the NRC (1984) guidelines for all essential amino acids. However, Han *et al.* (1992) determined that a 19% CP-corn-soybean meal diet is also deficient in Val. A Val deficiency may have contributed to the apparent Met imbalance and consequently lower performance of birds fed 18% CP.

Excess SAA has been associated with increased urinary Ca in humans (Zemel *et al.*, 1981) and rats (Whiting and Draper, 1981) and increased leg abnormalities in turkey poults (Stevens and Salmon, 1989). Excess CP (Stevens and Salmon, 1988) and excess total dietary amino acids (Skinner *et al.*, 1991) have been associated with reduced bone ash in turkey poults and broiler chicks, respectively. In contrast, results from the present study show that changes in tibia ash were more related to growth rate rather than CP or SAA levels *per se*. Calcium level (.6% Ca) was more

TABLE 6. Effect of protein level, calcium level, and sulfur amino acid levels on tibia variables and calcium excretion, Experiment 2

Treatment			Tibia			Ca excretion ⁴
Protein (P)	Ca	SAA	Ca ¹	Ash ²	TDW ³	
	(%)			(%)	(g)	(%)
18	1.0	.93	36.9	42.6 ^a	2.12 ^a	50.4 ^{ab}
18	1.0	1.23	37.4	43.4 ^a	2.09 ^{ab}	54.2 ^a
18	1.0	1.53	37.6	42.9 ^a	1.95 ^{abcd}	47.4 ^b
18	.6	.93	37.1	40.6 ^b	1.85 ^{cde}	35.7 ^c
18	.6	1.23	36.1	40.3 ^{bc}	1.87 ^{bcde}	31.5 ^c
18	.6	1.53	37.0	40.8 ^b	1.66 ^e	33.2 ^c
22	1.0	.93	36.8	43.0 ^a	2.15 ^a	54.5 ^a
22	1.0	1.23	36.2	42.3 ^a	2.14 ^a	54.6 ^a
22	1.0	1.53	38.5	42.4 ^a	2.04 ^{abc}	45.5 ^b
22	.6	.93	36.9	39.7 ^{bcd}	1.82 ^{cde}	36.3 ^c
22	.6	1.23	36.3	39.1 ^d	1.80 ^{de}	34.6 ^c
22	.6	1.53	36.1	39.3 ^{cd}	1.86 ^{bcde}	36.0 ^c
Pooled SEM			.78	.36	.07	1.8
Source of variation		df	Probability			
P		1	.640	<.001	.287	.167
Ca		1	.175	<.001	<.001	<.001
SAA		2	.368	.765	.073	.014
P × Ca		1	.821	.054	.755	.548
P × SAA		2	.906	.169	.221	.751
Ca × SAA		2	.337	.570	.805	.003
P × Ca × SAA		2	.351	.440	.505	.278
Contrasts						
SAA linear		1	.507	.616	.037	.007
SAA quadratic		1	.212	.596	.342	.233

^{a-e}Values within the same column with no common superscripts differ significantly ($P < .05$).

¹Average percentage tibia calcium of three birds per replication and six replications per treatment.

²Average tibia ash percentage of three birds per replication and six replications per treatment.

³Average weight of dried and fat-free tibia removed from three birds per replication.

⁴Calculated as Ca (grams) in feces over Ca (grams) consumption in a 48-h period. Values represent the average of six replications per treatment.

detrimental for tibia ash in birds fed 22% CP compared with birds fed 18% CP, indicating that birds with greater growth rate are more susceptible to low Ca levels for adequate bone mineralization. Although the 18% CP diet was supplemented with Lys and Arg, furnished as hydrochlorides, the acid load provided by the anion fraction of these synthetic amino acids did not impair bone mineralization. In fact, birds fed 18% CP had significantly higher tibia ash percentage than birds fed 22% CP.

Tibia weight decreased with increasing SAA, an effect apparently related to a decrease in growth rate and not to an increase in the catabolism of SAA. This was confirmed by the fact that SAA did

not have a significant effect on percentage tibia ash or percentage tibia Ca. Moreover, as SAA increased, Ca excretion in the feces decreased significantly. The lack of significant interactions between SAA and CP levels for tibia variables and the greater tibia ash percentage in birds fed 18% CP compared with birds fed 22% CP indicate that the acid load that might have been induced by supplementing the 18% CP diet with Lys and Arg as hydrochlorides did not affect bone mineralization.

The results of the present study, as with similar studies in other species, indicate that dietary inorganic sulfate above .034% may adversely affect bone mineralization of 3-wk-old broilers. Moreover, the effect of sulfate on bone ash depends on the

source of sulfate. However, in contrast to the findings of others, the levels of SAA used in the present study had no significant effect on percentage tibia ash or percentage tibia Ca. According to the acid-base balance concept of Mongin (1981), it would be expected that sulfate produced from excess Met or Cys would be more conducive to hypercalciuria through acidosis than an equivalent (or even greater) quantity of sulfur provided as a sulfate salt with a nonmetabolizable cation (e.g., potassium sulfate). To accurately define this apparent controversy, examination of the effect of sulfate on fecal and urinary Ca and blood acid-base balance is necessary. The present study indicates that SAA supplementation up to 65% in excess of the requirement does not adversely affect bone mineralization in 3-wk-old broilers, and the effects of CP level on tibia variables were more related to growth rate than Ca deposition.

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